

RESEARCH INTEREST

Long's research focuses on advancing optimization techniques for multi-objective deep neural networks. He specifically investigates optimization methods to improve model robustness, safety, and helpfulness simultaneously. A key area of his work involves scaling these effectiveness measures to large-scale applications, particularly Large Language Models (LLMs).

EDUCATION

PhD	Singapore University of Technology and Design (SUTD)	09/2024 – Present
- Academic advisors: Asst. Prof. Wenxuan Zhang and Prof. Wei Lu. - Major: Artificial Intelligence.		
Bachelor	Hanoi University of Science and Technology (HUST)	08/2018 – 09/2022
- Academic advisor: Dr. Thang N. Tran - Major: Mathematics and Informatics. CPA: 3.11/4.00 (Rank 18 out of 94 in his department).		

SELECTED PUBLICATIONS

- [1] **Long P. Hoang**, Wenxuan Zhang, Wei Lu, [Difficulty–Diversity Collaborative Filtering for Data-Efficient LLM Fine-Tuning](#), *In Proceedings of The International Conference on Learning Representations (ICLR)*, 2026.
- [2] Minh-Duc Nguyen, Phuong Mai Dinh, Quang-Huy Nguyen, **Long P. Hoang**, Dung D. Le, [Improving Pareto Set Learning for Expensive Multi-objective Optimization via Stein Variational Hypernetworks](#), *In Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.
- [3] Tuan A. Tran[†], **Long P. Hoang**[†], Dung D. Le, Thang N. Tran, [A Framework for Controllable Pareto Front Learning with Completed Scalarization Functions and its Applications](#), *Neural Networks*, 2024.
- [4] **Long P. Hoang**, Dung D. Le, Tuan A. Tran, Thang N. Tran, [Improving Pareto Front Learning via Multi-Sample Hypernetworks](#), *In Proceedings of the AAAI Conference on Artificial Intelligence*, 2023 (Oral).

Preprints

- [1] **Long P. Hoang**[†], Hai V. Le[†], Shaoyang Xu, Wei Lu, Wenxuan Zhang, [Safety Paradox: How Enhanced Safety Awareness Leaves LLMs Vulnerable to Posterior Attack](#), *arXiv preprint arXiv:2606.05614*, 2026.
- [2] **Long P. Hoang**, Yiran Zhao, Wei Lu, Wenxuan Zhang, [Multilingual Fine-Tuning via Localized Gradient Conflict Resolution](#), *arXiv preprint arXiv:2606.05613*, 2026.

EXPERIENCE

PhD Student	The inclusive NLP Lab
Advisor: Prof. Wei Lu and Asst. Prof. Wenxuan Zhang	
<i>How Enhanced Safety Awareness Leaves LLMs Vulnerable to Posterior Attack</i>	10/2025 – 05/2026

[†]Co-First Author

- Developed Posterior Attack, a single-query jailbreak method that demonstrates high efficacy against a wide range of LLMs, including GPT-5 and Claude 4.6.
- Using Bayesian inference, developed a theoretical framework showing the Safety Paradox between LLM internal safety-awareness and its vulnerability under Posterior Attack (submitted to EMNLP 2026).

Multilingual Fine-Tuning via Localized Gradient Conflict Resolution

01/2026 – 05/2026

- Formulated multilingual fine-tuning as a multi-objective optimization (MOO) problem, then developed distributed gradient-based MOO training to improve LLM performance up to 4% (submitted to NeurIPS 2026).

Difficulty-Diversity Collaborative Filtering for Data-Efficient LLM Fine-Tuning

04/2025 – 12/2025

- Developed a collaborative filtering framework to select a small, challenging, and diverse dataset personalized for a given language model (accepted to ICLR 2026).
- Empirically validated that fine-tuning with 1,000 high-quality examples selected via our method yields better performance than using the entire 220,000-example dataset.

Research Assistant

College of Engineering and Computer Science, VinUniversity

Advisor: [Asst. Prof. Dung D. Le](#)

Expensive Multi-Objective Optimization

03/2023 – 10/2023

- Approximated the entire trade-off curve of black-box objects using Pareto Front Learning with Hypernetworks computed by Gaussian Processes (accepted to AAAI 2025).

Profiling the Pareto Front in Multi-Task Learning

02/2022 – 10/2022

- Proposed a novel method named Multi-Sample Hypernetwork to approximate the entire trade-off curve of conflicting objectives (accepted to AAAI 2023).

Undergraduate Research Assistant

School of Applied Mathematics and Informatics, HUST

Advisor: [Dr. Tran Ngoc Thang](#)

Multi-Objective Optimization with Completed Scalarizations

02/2022 – 02/2023

- Proposed and proved the convergence of a new framework for Pareto Multi-Task Learning with Scalarization Functions in the pseudo-convex and quasiconvex assumptions (accepted to the journal Neural Networks).

ADWARDS & CERTIFICATES

- AISG PhD Fellowship, granted by AI Singapore, Sep 2024 – Sep 2028.
- Certificate of Completion of Developer Circles Vietnam Innovation Challenge in Data Science, sponsored by Facebook, Nov 2020.

REFERENCES

1. [Assistant Professor Wenxuan Zhang](#) (Ph.D), Information Systems Technology and Design, Singapore University of Technology and Design wxzhang@sutd.edu.sg
2. [Professor Wei Lu](#) (Ph.D), College of Computing and Data Science, Nanyang Technological University wei.lu@ntu.edu.sg
3. [Assistant Professor Dung D. Le](#) (Ph.D), College of Engineering and Computer Science, VinUniversity dung.ld@vinuni.edu.vn
4. [Dr. Thang N. Tran](#) (Ph.D), Faculty of Applied Mathematics and Informatics, Hanoi University of Science and Technology thang.tranngoc@hust.edu.vn